

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

36863

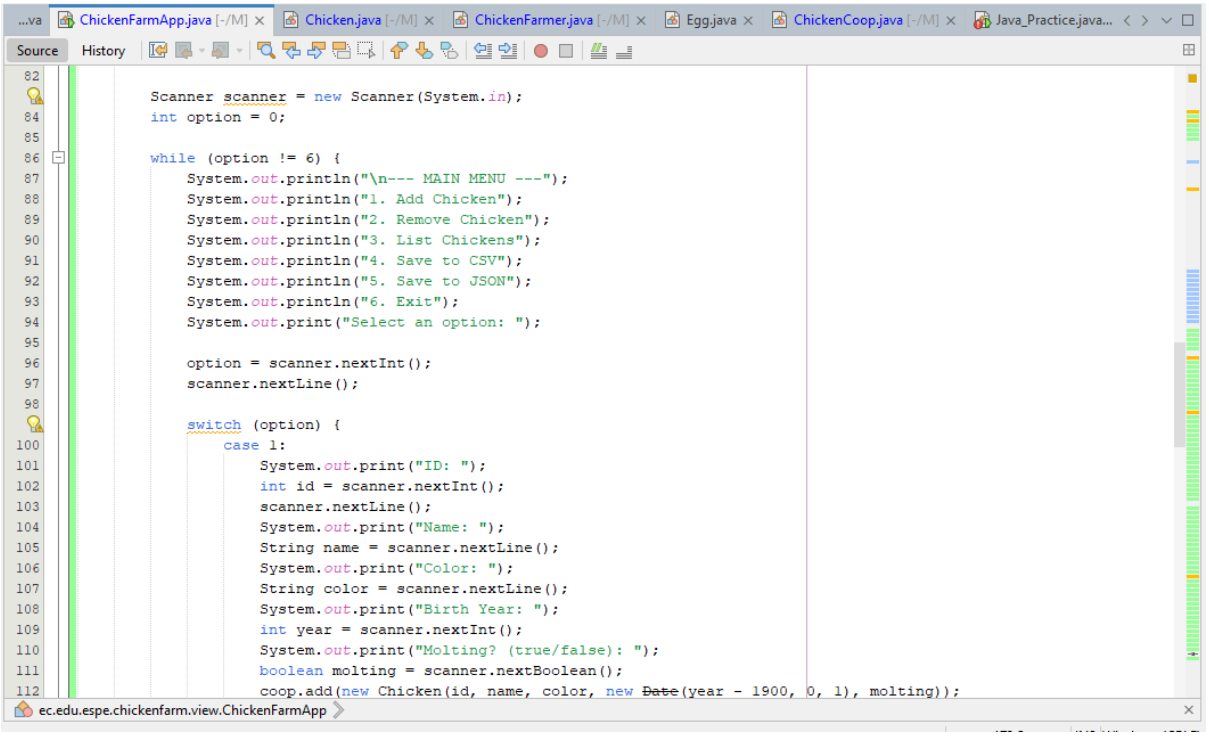
Homework09 - ChickenFarmAppCreateandRead

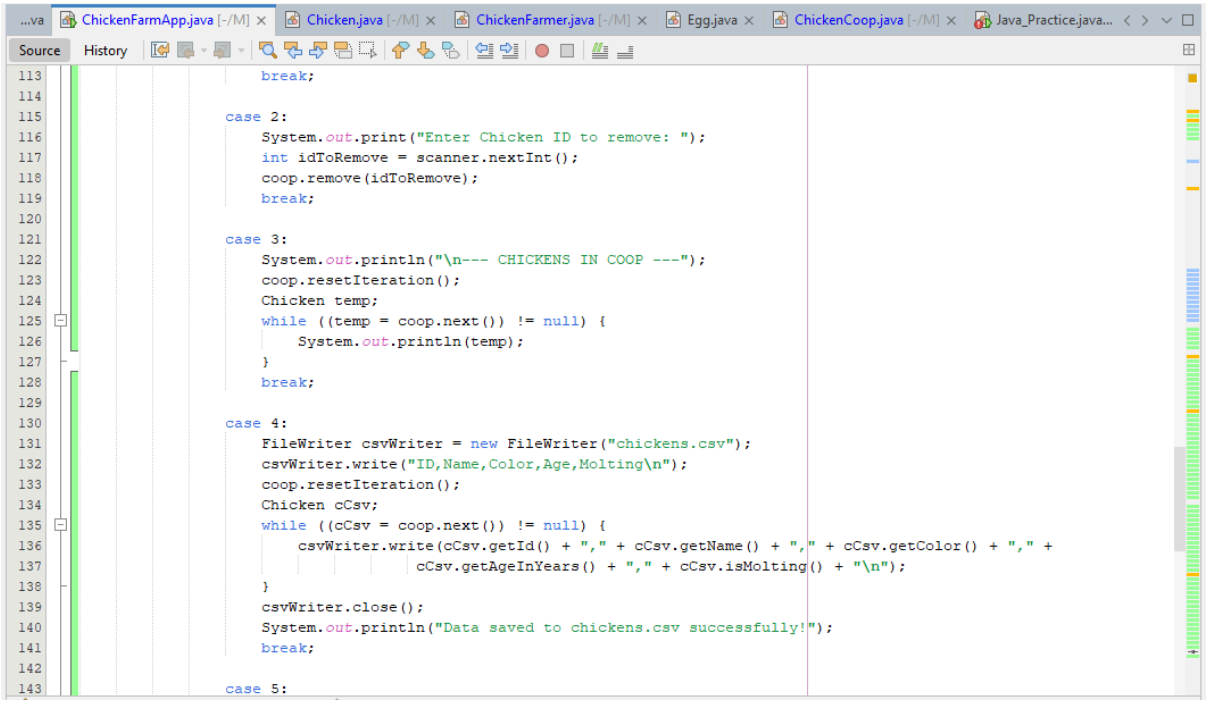
LASCANO, JORGE EDISON

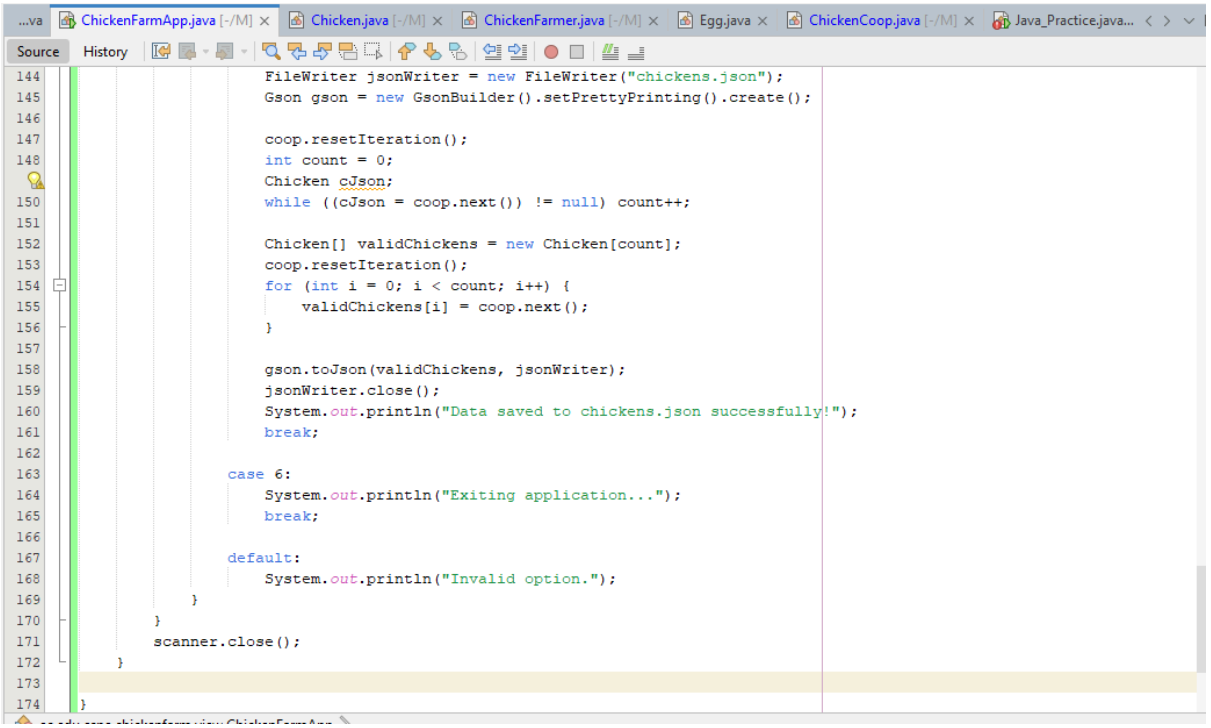
October 14, 2026

Evidence

Java:







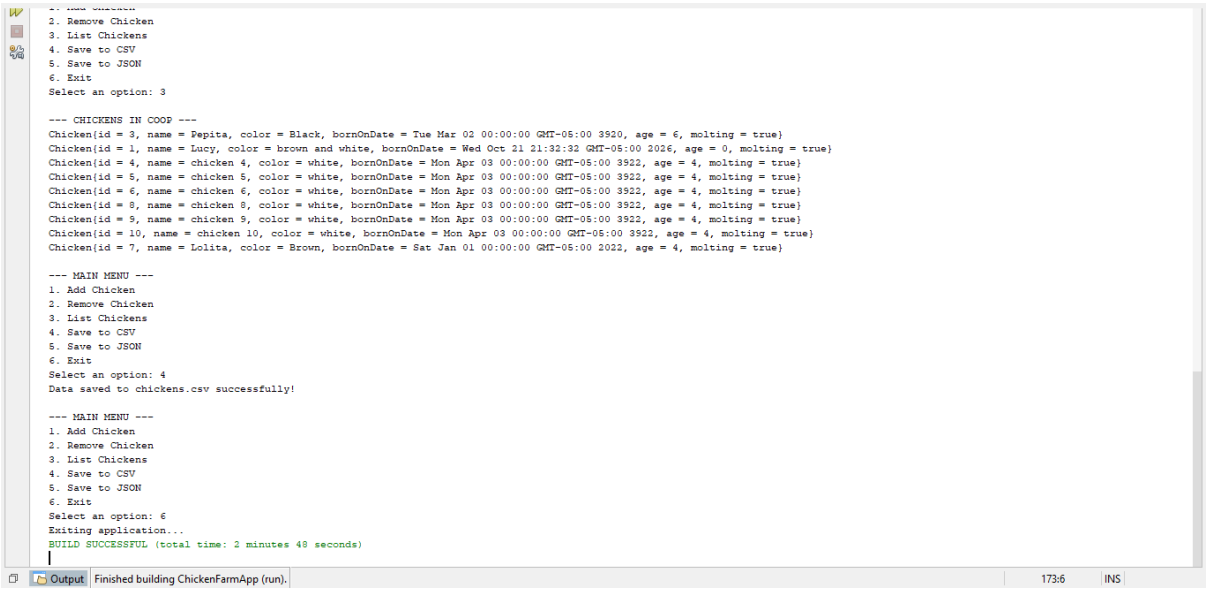
```
...va | ChickenFarmApp.java [-/M] | Chicken.java [-/M] | ChickenFarmer.java [-/M] | Egg.java | ChickenCoop.java [-/M] | Java_Practice.java...
Source  History
144                     FileWriter jsonWriter = new FileWriter("chickens.json");
145                     Gson gson = new GsonBuilder().setPrettyPrinting().create();
146
147                     coop.resetIteration();
148                     int count = 0;
                        Chicken cJson;
150                     while ((cJson = coop.next()) != null) count++;
151
152                     Chicken[] validChickens = new Chicken[count];
153                     coop.resetIteration();
154                     for (int i = 0; i < count; i++) {
155                         validChickens[i] = coop.next();
156                     }
157
158                     gson.toJson(validChickens, jsonWriter);
159                     jsonWriter.close();
160                     System.out.println("Data saved to chickens.json successfully!");
161                     break;
162
163                 case 6:
164                     System.out.println("Exiting application...");
165                     break;
166
167                 default:
168                     System.out.println("Invalid option.");
169             }
170         }
171         scanner.close();
172     }
173
174 }
```

```
--- MAIN MENU ---
1. Add Chicken
2. Remove Chicken
3. List Chickens
4. Save to CSV
5. Save to JSON
6. Exit
Select an option: 1
ID: 7
Name: Lolita
Color: Brown
Birth Year: 2022
Molting? (true/false): true
Successfully added chicken: Lolita

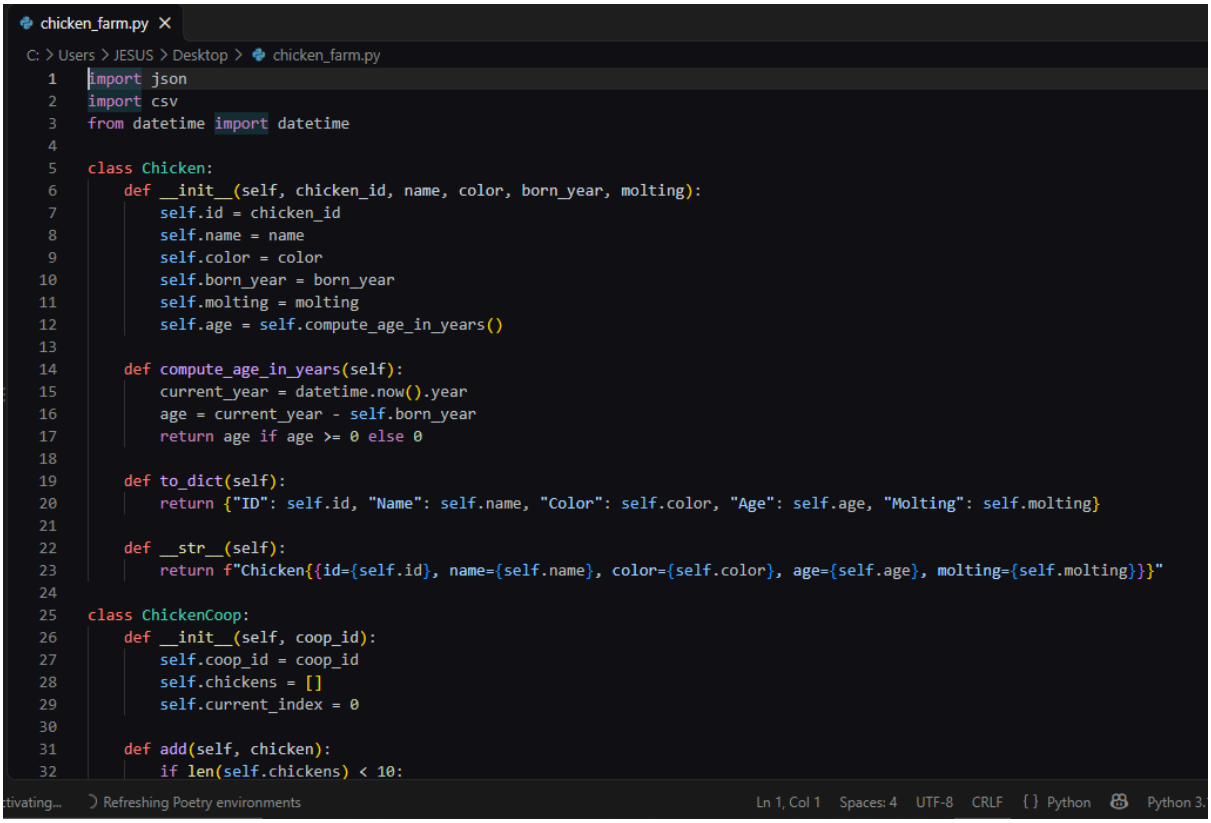
--- MAIN MENU ---
1. Add Chicken
2. Remove Chicken
3. List Chickens
4. Save to CSV
5. Save to JSON
6. Exit
Select an option: 2
Enter Chicken ID to remove: 7
Chicken with ID 7 has been removed.

--- MAIN MENU ---
1. Add Chicken
2. Remove Chicken
3. List Chickens
4. Save to CSV
5. Save to JSON
6. Exit
Select an option: 3

--- CHICKENS IN COOP ---
Chicken{id = 3, name = Pepita, color = Black, bornOnDate = Tue Mar 02 00:00:00 GMT-05:00 3920, age = 6, molting = true}
Chicken{id = 1, name = Lucy, color = brown and white, bornOnDate = Wed Oct 21 21:32:32 GMT-05:00 2026, age = 0, molting = true}
Chicken{id = 4, name = chicken 4, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
```


2. Remove Chicken
3. List Chickens
4. Save to CSV
5. Save to JSON
6. Exit
Select an option: 3
--- CHICKENS IN COOP ---
Chicken{id = 3, name = Pepita, color = Black, bornOnDate = Tue Mar 02 00:00:00 GMT-05:00 3920, age = 6, molting = true}
Chicken{id = 1, name = Lucy, color = brown and white, bornOnDate = Wed Oct 21 21:32:32 GMT-05:00 2026, age = 0, molting = true}
Chicken{id = 4, name = chicken 4, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
Chicken{id = 5, name = chicken 5, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
Chicken{id = 6, name = chicken 6, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
Chicken{id = 8, name = chicken 8, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
Chicken{id = 9, name = chicken 9, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
Chicken{id = 10, name = chicken 10, color = white, bornOnDate = Mon Apr 03 00:00:00 GMT-05:00 3922, age = 4, molting = true}
Chicken{id = 7, name = Lolita, color = Brown, bornOnDate = Sat Jan 01 00:00:00 GMT-05:00 2022, age = 4, molting = true}
--- MAIN MENU ---
1. Add Chicken
2. Remove Chicken
3. List Chickens
4. Save to CSV
5. Save to JSON
6. Exit
Select an option: 4
Data saved to chickens.csv successfully!
--- MAIN MENU ---
1. Add Chicken
2. Remove Chicken
3. List Chickens
4. Save to CSV
5. Save to JSON
6. Exit
Select an option: 6
Exiting application...
BUILD SUCCESSFUL (total time: 2 minutes 40 seconds)
Output
Finished building ChickenFarmApp (run).
173:6
INS

Python:


chicken_farm.py
C: > Users > JESUS > Desktop > chicken_farm.py
import json
import csv
from datetime import datetime
class Chicken:
def __init__(self, chicken_id, name, color, born_year, molting):
self.id = chicken_id
self.name = name
self.color = color
self.born_year = born_year
self.molting = molting
self.age = self.compute_age_in_years()
def compute_age_in_years(self):
current_year = datetime.now().year
age = current_year - self.born_year
return age if age >= 0 else 0
def to_dict(self):
return {"ID": self.id, "Name": self.name, "Color": self.color, "Age": self.age, "Molting": self.molting}
def __str__(self):
return f"Chicken{{id={self.id}, name={self.name}, color={self.color}, age={self.age}, molting={self.molting}}}"
class ChickenCoop:
def __init__(self, coop_id):
self.coop_id = coop_id
self.chickens = []
self.current_index = 0
def add(self, chicken):
if len(self.chickens) < 10:
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Refreshing Poetry environments
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Python 3.

chicken_farm.py

C: > Users > JESUS > Desktop > chicken_farm.py

```python
def main():
    coop = ChickenCoop(1)
    coop.add(Chicken(2, "Maruja", "White", 1970, False))
    coop.add(Chicken(1, "Lucy", "Brown and white", datetime.now().year, True))

    while True:
        print("\n--- MAIN MENU ---")
        print("1. Add Chicken\n2. Remove Chicken\n3. List Chickens\n4. Save to CSV\n5. Save to JSON\n6. Exit")
        option = input("Select an option: ")

        if option == '1':
            id_val = int(input("ID: "))
            name = input("Name: ")
            color = input("Color: ")
            year = int(input("Birth Year: "))
            molting = input("Molting? (true/false): ").lower() == 'true'
            coop.add(Chicken(id_val, name, color, year, molting))

        elif option == '2':
            id_to_remove = int(input("Enter Chicken ID to remove: "))
            coop.remove(id_to_remove)

        elif option == '3':
            print("\n--- CHICKENS IN COOP ---")
            coop.reset_iteration()
            while True:
                c = coop.next()
                if c is None: break
                print(c)

        elif option == '4':
            with open('chickens.csv', 'w', newline='') as file:
```

ady Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14 (64-bit)

C: > Users > JESUS > Desktop > chicken_farm.py

```python
def main():

        elif option == '3':
            print("\n--- CHICKENS IN COOP ---")
            coop.reset_iteration()
            while True:
                c = coop.next()
                if c is None: break
                print(c)

        elif option == '4':
            with open('chickens.csv', 'w', newline='') as file:
                writer = csv.writer(file)
                writer.writerow(["ID", "Name", "Color", "Age", "Molting"])
                for c in coop.chickens:
                    writer.writerow([c.id, c.name, c.color, c.age, c.molting])
            print("Data saved to chickens.csv successfully!")

        elif option == '5':
            with open('chickens.json', 'w') as file:
                json.dump([c.to_dict() for c in coop.chickens], file, indent=4)
            print("Data saved to chickens.json successfully!")

        elif option == '6':
            print("Exiting application...")
            break
        else:
            print("Invalid option.")

if __name__ == "__main__":
    main()
```

JavaScript

chickenFarm.js

C: > Users > JESUS > Desktop > chickenFarm.js > ...

```
const fs = require('fs');
const readline = require('readline');

class Chicken {
    constructor(id, name, color, bornYear, molting) {
        this.id = id;
        this.name = name;
        this.color = color;
        this.bornYear = bornYear;
        this.molting = molting;
        this.age = this.computeAgeInYears();
    }

    computeAgeInYears() {
        const currentYear = new Date().getFullYear();
        let age = currentYear - this.bornYear;
        return age >= 0 ? age : 0;
    }

    toString() {
        return `Chicken{id=${this.id}, name=${this.name}, color=${this.color}, age=${this.age}, molting=${this.molting}}`;
    }
}

class ChickenCoop {
    constructor(id) {
        this.id = id;
        this.chickens = [];
        this.currentIndex = 0;
    }

    add(chicken) {
```

chickenFarm.js

C: > Users > JESUS > Desktop > chickenFarm.js > ...

```
class ChickenCoop {
    add(chicken) {
        } else {
            console.log("Error: The chicken coop is completely full!");
        }
    }

    remove(chickenId) {
        const index = this.chickens.findIndex(c => c.id === chickenId);
        if (index !== -1) {
            this.chickens.splice(index, 1);
            console.log(`Chicken removed --> ${chickenId}`);
        } else {
            console.log("Chicken not found");
        }
    }

    resetIteration() {
        this.currentIndex = 0;
    }

    next() {
        if (this.currentIndex < this.chickens.length) {
            return this.chickens[this.currentIndex++];
        }
        return null;
    }
}

const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout
```

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chickenFarm.js

C: > Users > JESUS > Desktop > chickenFarm.js > ...

```
69  async function main() {
71      coop.add(new Chicken(2, "Maruja", "White", 1970, false));
72      coop.add(new Chicken(1, "Lucy", "Brown and white", new Date().getFullYear(), true));
73
74      let option = '0';
75      while (option !== '6') {
76          console.log("\n--- MAIN MENU ---");
77          console.log("1. Add Chicken\n2. Remove Chicken\n3. List Chickens\n4. Save to CSV\n5. Save to JSON\n6. Exit");
78          option = await question("Select an option: ");
79
80          switch (option) {
81              case '1':
82                  const id = parseInt(await question("ID: "));
83                  const name = await question("Name: ");
84                  const color = await question("Color: ");
85                  const year = parseInt(await question("Birth Year: "));
86                  const molting = (await question("Molting? (true/false): ")).toLowerCase() === 'true';
87                  coop.add(new Chicken(id, name, color, year, molting));
88                  break;
89              case '2':
90                  const idToRemove = parseInt(await question("Enter Chicken ID to remove: "));
91                  coop.remove(idToRemove);
92                  break;
93              case '3':
94                  console.log("\n--- CHICKENS IN COOP ---");
95                  coop.resetIteration();
96                  let c;
97                  while ((c = coop.next()) !== null) {
98                      console.log(c.toString());
99                  }
100                 break;
101             case '4':
```

chickenFarm.js

C: > Users > JESUS > Desktop > chickenFarm.js > main

```
69  async function main() {
95                  coop.resetIteration();
96                  let c;
97                  while ((c = coop.next()) !== null) {
98                      console.log(c.toString());
99                  }
100                 break;
101             case '4':
102                 let csvContent = "ID,Name,Color,Age,Molting\n";
103                 coop.chickens.forEach(chicken => {
104                     csvContent += `${chicken.id},${chicken.name},${chicken.color},${chicken.age},${chicken.molting}\n`;
105                 });
106                 fs.writeFileSync('chickens.csv', csvContent);
107                 console.log("Data saved to chickens.csv successfully!");
108                 break;
109             case '5':
110                 fs.writeFileSync('chickens.json', JSON.stringify(coop.chickens, null, 4));
111                 console.log("Data saved to chickens.json successfully!");
112                 break;
113             case '6':
114                 console.log("Exiting application...");
115                 rl.close();
116                 break;
117             default:
118                 console.log("Invalid option.");
119         }
120     }
121 }
122
123 main();
```